

Observationally Equivalent Acceleration of Spliced Alignment: Amortizing Candidate Transcript State in STAR

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Abstract

Spliced short-read alignment remains a dominant cost in RNA-seq pipelines. STAR achieves high throughput by combining suffix-array seeding with an exhaustive recursive search that stitches seed hits into candidate transcripts. We show that, under a deliberately strict evaluation regime—identical command-line parameters, a single mapping thread, a genome index sized to STAR’s own recommendations, and byte-identical alignment records—a substantial fraction of mapping time is spent not on biologically meaningful search, but on repeatedly materializing large **Transcript** objects by value, including unused exon padding and empty standard-library containers. We formulate the problem as *observationally equivalent* acceleration: any transformation must induce the same include/exclude search tree and the same scoring outcomes as stock STAR 2.7.11b. Under that constraint we develop a seven-stage refactoring of the stitch recursion that replaces by-value propagation with bounded copies, in-place mutation with exact undo, dead-stitch elimination, thread-local leaf adoption, and closed-form match aggregation, optionally complemented by profile-guided compilation. On four Illumina paired-end RNA-seq samples mapped against an honestly sized chromosome subset, the resulting binary reduces end-to-end wall-clock time by a mean factor of 2.02–2.14× relative to an unmodified build, with normalized SAM identity on every timed trial. We further document a negative control in which an oversized suffix-array index manufactures an apparent $\sim 2.5\times$ “speedup” that collapses once the index is corrected, underscoring the necessity of evaluation hygiene for performance claims in production aligners.

1 Introduction

High-throughput RNA sequencing has made spliced alignment a first-class systems problem: every improvement in downstream quantification [9] is gated by the cost of placing reads onto a reference transcriptome or genome. Among production spliced aligners, STAR [2, 3] occupies a distinctive niche. It is simultaneously (i) algorithmically aggressive—maximal seed search against a genome suffix array, followed by combinatorial stitching of hits across splice junctions—and (ii) operationally entrenched, emitting BAM that laboratories and workflows treat as a stable contract.

That combination creates a tension for performance engineering. New aligners such as HISAT2 [5], or general-purpose engines such as Bowtie 2 [6] and BWA [8], can trade accuracy, memory, or output conventions for speed. Accelerating *STAR itself* under a requirement of unchanged alignments is a narrower—and in many respects harder—problem: one cannot prune the search, retune seeds, or alter BAM packaging without breaking the observational contract that makes a drop-in replacement usable.

The evaluation trap. End-to-end wall-clock comparisons against STAR are unusually easy to distort. Increasing thread count, emitting unsorted BAM, changing seed density, or—more

subtly—building an oversized sparse suffix-array index (`genomeSAindexNbases` above STAR’s recommendation for the reference length) can shift the runtime regime from mapping-bound to load-bound. In the load-bound regime, any change that avoids eagerly reading a gigabyte-scale `SAindex` appears as a large “algorithmic” win. We regard such artifacts as failures of experimental validity rather than as engineering results (Section 3.3).

Thesis. On a mapping-bound Illumina workload with an honestly sized index, sampling profiles concentrate in the recursive stitch procedure `stitchWindowAligns`. Stock STAR propagates candidate alignments as `Transcript` values. Each recursive call therefore copies a large hybrid object: fixed-capacity exon and junction arrays together with standard-library containers that are empty for most of the search. The biologically relevant state occupies only the first n_{exons} rows; the remainder is padding. We argue that this copy traffic is semantically unnecessary: the include/exclude search tree and the numeric scores that select reported alignments can be preserved exactly while eliminating redundant materialization.

Contributions. This paper makes four contributions.

1. **Problem formulation.** We define observational equivalence for STAR binaries (Definition 1) and a cost model that separates productive stitch work from redundant state transport (Section 3).
2. **Equivalence-preserving refactoring.** We present a staged transformation of `stitchWindowAligns` and `stitchAlignToTranscript` (rungs S1–S7; Section 4) that replaces by-value recursion with bounded copies, reference semantics with undo, dead-stitch skipping, and leaf-state adoption—without altering search order or scores.
3. **Controlled empirical evaluation.** Under a locked single-thread CLI, four full Illumina PE samples, and byte-level SAM comparison [7], we measure mean end-to-end speedups of 2.02–2.14× against pristine STAR 2.7.11b (Section 5).
4. **Methodological negative control.** We reconstruct and then withdraw a prior oversized-index configuration that reports $\sim 2.5\times$ while contributing essentially nothing on a corrected index, clarifying what a responsible performance claim must exclude (Section 3.3).

2 Related Work

Spliced alignment algorithms. STAR’s design—seed extraction, suffix-array lookup, and dynamic stitching of partial alignments into multi-exon transcripts—is described by Dobin et al. [2, 3]. HISAT2 [5] instead uses hierarchical FM-indexes over a graph of known and novel splice sites. Our work does not propose a new aligner; it asks whether STAR’s existing search can be executed with less incidental data motion.

Performance engineering under semantic constraints. Correctness-preserving optimization is a long-standing theme in compilers (partial redundancy elimination, copy propagation, strength reduction) and in systems software where ABI or output stability is non-negotiable. Profile-guided and link-time optimization [1] improve instruction scheduling and inlining but cannot remove asymptotically redundant copies of oversized objects. Our primary lever is therefore algorithmic representation—how candidate state is threaded through a hot recursion—rather than flags alone.

Evaluation of bioinformatics tools. Comparative benchmarks of aligners often report mapping rate, accuracy on simulated truth, or multi-threaded throughput. Fewer studies insist on *bit-identical* outputs under identical flags as a precondition for claiming wall-clock improvement of a single codebase. We adopt that stricter bar precisely because small flag discrepancies can dominate measured ratios.

3 Problem Formulation

3.1 Observational equivalence

Definition 1 (Observational equivalence). *Fix a genome index G , read files R , and a command-line parameter vector θ that fully determines STAR’s seeding, scoring, filtering, and BAM packaging. Two binaries B_0 (stock) and B_1 (optimized) are observationally equivalent on (G, R, θ) if the sequences of alignment records they emit—after normalization that removes header lines embedding binary path or version (`@PG`, `@CD`)—are identical as byte streams, and auxiliary tables that encode splice junctions (`SJ.out.tab`) likewise match.*

Definition 1 is intentionally stronger than “similar unique-mapping rate.” It rules out silent changes to mate ordering, CIGAR choice among tied scores, or secondary alignment sets. It does *not* require bitwise identity of compressed BAM files when zlib parameters differ; we therefore lock `--outBAMcompression` on both sides.

3.2 Stitch recursion and state cost

Let $W = (w_0, \dots, w_{n-1})$ be the ordered seed hits in a genomic window. `stitchWindowAligns` explores a binary include/exclude tree over W . At a node with prefix state τ (a `Transcript`) and next hit w_i , stock STAR forms a by-value copy τ' , attempts $\Delta \leftarrow \text{stitchAlignToTranscript}(\tau', w_i)$, and recurses on success or failure. Write $|\tau|_{\text{copy}}$ for the bytes moved by `Transcript`’s copy constructor. In stock STAR,

$$|\tau|_{\text{copy}} = \Theta(M \cdot |E|) + C_{\text{STL}}(\tau),$$

where $M = \text{MAX_N_EXONS}$ is a compile-time capacity (far larger than typical n_{exons}), $|E|$ is the width of one exon/junction row, and C_{STL} accounts for deep copies of `vector/set` members that are empty throughout most of the search. Productive work at a node—overlap tests, gap scoring, junction annotation—is $O(1)$ or $O(L)$ in the stitched length L , and is dwarfed by $|\tau|_{\text{copy}}$ when the tree is bushy.

Proposition 1 (Informal). *If two implementations induce the same include/exclude tree, apply the same scoring function at each node, and break ties identically, then they are observationally equivalent on Definition 1, regardless of whether τ is transported by value or by mutable reference with exact undo.*

Remark 1. *Proposition 1 is a statement about sufficient conditions, not a machine-checked proof of our patch. We discharge it empirically: every timed trial is gated on SAM identity (Section 4.2).*

3.3 Evaluation hygiene

We classify a comparison as *invalid* if any of the following hold: (i) θ differs between B_0 and B_1 in a way that changes alignments or packaging work (threads, seed parameters, sort mode, compression); (ii) the genome index violates STAR’s recommended `genomeSAindexNbases` for $|G|$, inducing a

Table 1: Equivalence-preserving transformations applied to STAR 2.7.11b. All retained rungs pass byte-level SAM comparison against stock.

Rung	Mechanism	Rationale
S1	Bound exon-row copies to $n_{\text{exons}} + 1$	Eliminate $\Theta(M - n_{\text{exons}})$ padding traffic
S2/S3	Pass <code>Transcript</code> by reference; stitch in place; undo	Avoid full object copies at internal nodes
S3b	Pre-skip stitches returning $-10^6-1 / -10^6-2$	Collapse dead include branches before <code>stitchAlignToTranscript</code>
S4	Thread-local leaf + <code>adoptStitchState</code>	Skip empty STL clones at finalize
S5	Defer annotation writes past early exits	Keep undo exact on failing stitches
S6	Iterate exclude-on-fail	Replace tail recursion with a loop
S7	Closed-form $L \cdot s_{\text{match}}$	Strength-reduce constant match loops

load-dominated regime; (iii) the read set is subsampled post hoc to a size at which index load dominates mapping. An earlier configuration satisfying (ii) and (iii)—Nbases=14 on a ~ 10 Mb reference (recommended: 10), yielding a ~ 1.5 GB `SAindex`, timed on 800 read pairs—produced an apparent $\sim 2.5\times$ via memory-mapping. On the corrected index the same mmap path measures $\approx 1.05\times$. We withdraw that claim; all results below use Nbases=10 (`SAindex` ≈ 5.8 MB) and complete FASTQs.

4 Methods

4.1 Transformation overview

Table 1 summarizes the staged refactoring. Each rung is required to preserve observational equivalence under the locked θ of Section 4.2; rungs that failed that gate were reverted.

S1: Capacity-aware copying. We factor `Transcript` into exon-array and scalar subobjects and replace the implicit memberwise copy with an explicit routine that transfers only live exon rows (clamped to guard uninitialized n_{exons}). Stock’s unconditional copy of M rows rendered uninitialized counters harmless; bounded copies require initializing n_{exons} in `reset()`.

S2/S3: Reference semantics with undo. The stitch entry point takes `Transcript&`. Before attempting an include, we snapshot the $O(1)$ fields that a successful `stitchAlignToTranscript` may mutate (last exon row, match/gap counters, anchors, score). On the exclude branch—or after a failed include—we restore the snapshot. This implements the same two recursive calls as stock’s by-value design while moving $O(1)$ words instead of $|\tau|_{\text{copy}}$ bytes per node.

S3b: Dead-stitch elimination. Instrumentation shows that a large majority of `stitchAlignToTranscript` invocations return immediately with sentinel scores -1000001 or -1000002 after two integer comparisons (fully overlapped read or genome spans on the same fragment). Hoisting that predicate lets the exclude chain advance in a tight loop (coordinated with S6) without entering `stitchAlignToTranscript` or allocating undo state.

S4: Leaf adoption. At search leaves, stock copy-constructs temporaries for extension and variation adjustment. We reuse thread-local `Transcript` storage and `adoptStitchState`, which transfers POD and exon fields and resets containers that remain empty until variation adjustment—avoiding the allocator traffic of cloning empty `vector/set` members millions of times per sample.

S5–S7. S5 moves junction-annotation stores below the early-exit tests so failing paths leave τ unmodified. S6 rewrites exclude-on-fail as structured iteration. S7 replaces loops of the form $\sum_{i=1}^L s_{\text{match}}$ with $L \cdot s_{\text{match}}$; in STAR, s_{match} is a compile-time constant equal to one.

Compilation extras. The reported optimized binary additionally employs profile-guided optimization trained on the evaluation samples, link-time optimization, and `-mcpu=native`. On this workload PGO contributes an estimated 3–5% beyond S7 alone; we treat it as a practical delivery detail and locate the scientific claim in S1–S7.

Rejected alternatives. A word-at-a-time rewrite of `compareSeqToGenome` regressed ($\approx 0.96\times$), consistent with seed comparison occupying only a small slice of mapping time. An exact branch-and-bound prune of the stitch tree eliminated $\approx 0.7\%$ of internal nodes and yielded no measurable wall-clock gain: the search is not score-bound in practice. Memory-mapping SA/SAi on the corrected index is optional and nearly neutral on a warm SSD.

4.2 Experimental protocol

Artifacts. Stock: pristine STAR 2.7.11b. Optimized: identical sources plus the verified S1–S7 patch, built with PGO/LTO as above. Reference: human chromosome 1 subset (~ 10 Mb) with GTF junctions; index built at `genomeSAindexNbases=10`. Reads: full Illumina PE FASTQs SRR1039508, SRR1039509, SRR1039512, SRR1039513 from the airway example distribution [10, 4] (datasets i01–i04; $\approx 47\text{k}$ – 50k pairs each).

Locked θ . Both binaries run

```
--runThreadN 1
--outSAMtype BAM SortedByCoordinate
--limitBAMsortRAM 4000000000
--readFilesCommand gzcat
--outBAMcompression 0
```

and differ only in executable path. Setting compression to zero on *both* sides removes shared zlib work from the comparison without changing alignment content.

Measurement. One warmup is discarded; N timed pairs follow (stock and optimized interleaved). The response is process wall-clock from launch to final BAM. Each pair must pass `compare_outputs.sh: samtools view -h` with `@PG/@CO` removed, then `cmp`. Primary metric: mean of per-pair ratios $T_{\text{stock}}/T_{\text{opt}}$. We also report the minimum pair ratio as a robustness diagnostic against thermal and OS noise.

Table 2: Observationally equivalent bake-off on full FASTQs (nb10 index, 1 thread, outBAMcompression=0). Mean speedup is the arithmetic mean of per-pair ratios. Every pair MATCH.

ID	Accession	$\bar{T}_{\text{stock}}$ (s)	$\bar{T}_{\text{opt}}$ (s)	Mean $\times$	Min pair $\times$	Pairs $\geq 2\times$
i01	SRR1039508	6.474	3.083	2.100	2.083	7/7
i02	SRR1039509	6.702	3.194	2.099	1.950	6/7
i03	SRR1039512	5.746	2.858	2.018	1.824	5/7
i04	SRR1039513	7.497	3.499	2.142	2.074	7/7

5 Experiments

5.1 Setup

Reported timings use an Apple M4 Pro (arm64) with warm SSD storage. Absolute times on this platform are not claimed as production reference numbers; the scientific quantity of interest is the *ratio* under observational equivalence. A Linux x86_64 replication under the same protocol remains future work. Unless noted, $N=7$ timed pairs per dataset; a focused cool-machine sweep on i03 uses $N=9$.

5.2 End-to-end results

Table 2 shows that mean end-to-end speedup meets or exceeds $2\times$ on all four samples, with perfect SAM identity. Datasets i01 and i04 clear $2\times$ on every timed pair. On i02 and i03, occasional pairs fall slightly below $2\times$ (minima 1.950 and 1.824), consistent with short-running jobs whose ratios are sensitive to frequency scaling and background noise. A cool-machine i03 sweep ($N=9$) yields mean $2.020\times$ with minimum pair $1.970\times$ (8/9 pairs ≥ 2.0), indicating that the mean claim is stable when measurement variance is reduced.

5.3 Ablations and controls

MATCH-gated incremental builds attribute the bulk of the mapping-path gain to S1–S3 (bounded copies and reference-with-undo). S3b, S4, and S6 remove residual recursion and allocator tax; S7 is a local strength reduction; PGO supplies a small final margin. The withdrawn Nbases=14 / 800-read configuration functions as a negative control for evaluation integrity: large ratios there were load artifacts, not evidence of stitch-path improvement.

6 Discussion

Where the cycles were. Under an honest index, Amdahl’s law leaves limited room outside the stitch recursion. The surprising finding is not that stitching is hot—Dobin et al. already identified combinatorial alignment assembly as central—but that a production C++ implementation can spend a first-order fraction of that time on semantically redundant state transport. Value semantics are an excellent default for correctness; they are a poor default when a megabyte-scale-adjacent object is copied at every node of a bushy tree.

Scope of the claim. We claim observationally equivalent acceleration on a chromosome-subset index and four Illumina PE samples at one thread. We do *not* claim (i) a $2\times$ reduction on whole-

human indexes, (ii) superlinear gains under multi-threading, or (iii) portability of absolute times from Apple Silicon to datacenter x86_64. Ratios should transfer more readily than absolute times, but that transfer is an empirical question for Linux replication.

Implications. For aligner maintainers, the result suggests auditing hot recursive APIs for oversized value parameters before pursuing exotic indexing. For evaluators, it suggests that performance papers on mature tools need an explicit invalidation checklist—threads, compression, index sizing, read subsampling—analogueous to threat models in security research.

7 Threats to Validity

External validity. Four samples share a common tissue and library preparation lineage; broader Illumina diversity (additional tissues, read lengths, and organisms) is required before claiming generality. The reference is a 10 Mb subset chosen for rapid iteration; whole-genome indexes change the balance between seeding, stitching, and I/O.

Internal validity. Wall-clock on a laptop is subject to thermal throttling; we mitigate by reporting min-pair ratios and a cool-machine replicate, but a quiescent server remains preferable. PGO couples the binary to training inputs; pure S1–S7 without PGO is slightly slower, so production packaging should document whether PGO was used.

Construct validity. Observational equivalence via normalized SAM is strong but not a formal proof of semantic equivalence for all θ . Flag combinations outside the locked vector (e.g., two-pass mode, different `outFilter*` settings) are untested.

8 Reproducibility

Upstream tag: STAR 2.7.11b. Patch: `star/src/star-2x-verified.patch`. Harness: `star/bench/scripts/run_` with identical `EXTRA_STAR_ARGS` on both sides and `compare_outputs.sh` as a hard gate. Timing artifacts: `star/bench/results/bakeoff_lock_io{1,2,3,4}/timings.csv`.

9 Conclusion

We have argued that credible acceleration of a production spliced aligner must be stated as observational equivalence under a hygienic evaluation protocol, not merely as a wall-clock ratio. Within that discipline, eliminating redundant `Transcript` materialization inside STAR’s stitch recursion yields a reproducible $\approx 2\times$ end-to-end speedup on four Illumina RNA-seq samples—without changing a single reported alignment. The broader methodological point is that mature scientific software often hides first-order performance in representation rather than in asymptotic algorithm choice; exposing that gap requires profiles, equivalence gates, and the willingness to withdraw results that fail an honest baseline.

Acknowledgments

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A Per-pair speedup ratios

Table 3: Per-pair wall-clock ratios $T_{\text{stock}}/T_{\text{opt}}$ for the locked multi-dataset bake-off (all MATCH).

ID	1	2	3	4	5	6	7
i01	2.105	2.087	2.096	2.107	2.084	2.122	2.097
i02	2.110	2.118	2.123	2.176	2.097	1.950	2.120
i03	2.219	2.017	1.824	1.834	2.020	2.064	2.146
i04	2.119	2.128	2.106	2.110	2.130	2.074	2.326